

## Problem Set 5: Surplus

### Questions and Solutions

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- 1) Consider a perfectly competitive market with no externalities. Market demand is given by  $Q^D = 65 - P$  while market supply is given by  $Q^S = P - 5$ . Calculate Consumer Surplus, Producer Surplus and Total Surplus.

**Solution:** We need to first find market quantity and price. We do that by setting

$$\begin{aligned} Q^D &= Q^S \\ 65 - P &= P - 5 \\ P &= 35 \\ \Rightarrow Q &= 30 \end{aligned}$$

To calculate CS we need the intercept of the demand curve on the y-axis, which is 65. This makes CS the area of the triangle with base = 30 (the quantity), height = 65-35 (the difference between price and the intercept of the demand curve on the y-axis).

$$CS = 0.5 * (65 - 35) * 30 = 450$$

Similarly, for PS we need the intercept of the supply curve on the y-axis, which is 5. This makes PS the area of the triangle with base = 30 (the quantity), height = 35-5 (the difference between price and the intercept of the supply curve on the y-axis).

$$PS = 0.5 * (35 - 5) * 30 = 450$$

Since we have no externalities, we can use the private benefits to calculate the social benefits. No members of society are affected by the buying and selling other than the consumers and producers, so we can calculate total surplus using

$$TS = CS + PS = 900$$

**Consumer surplus is \$450, producer surplus is \$450, and total surplus is \$900.**

- 2) We have a perfectly competitive market. Market demand is given by  $Q^D = 30 - \frac{1}{2}P$  while market supply is given by  $Q^S = 2P - 5$ .

**2a)** Calculate Consumer Surplus and Producer Surplus.

**Solution:** We proceed the same way as in the question above. We need to first find market quantity and price. We do that by setting

$$\begin{aligned} Q^D &= Q^S \\ 30 - \frac{1}{2}P &= 2P - 5 \\ P &= 14 \\ \Rightarrow Q &= 23 \end{aligned}$$

To calculate CS we need the intercept of the demand curve on the y-axis, which is 60. This makes CS the area of the triangle with base = 23 (the quantity), height = 60-14 (the difference between price and the intercept of the demand curve on the y-axis).

$$CS = 0.5 * (60 - 14) * 23 = 529$$

Similarly, for PS we need the intercept of the supply curve on the y-axis, which is 2.5. This makes PS the area of the triangle with base = 23 (the quantity), height = 14-2.5 (the difference between price and the intercept of the supply curve on the y-axis).

$$PS = 0.5 * (14 - 2.5) * 23 = 132.25$$

**Consumer surplus is \$529 and producer surplus is \$132.25.**

2b) Can you calculate total surplus?

**Solution:** Without being told that all benefits and costs accrue only to the parties in the transaction and no one outside (i.e. no externalities), we couldn't calculate total surplus. There could be benefits or costs accruing to people outside the transactions that would need to be counted in total surplus. Another way of saying this is that social benefit (or cost) could be different from private benefit (or cost). The demand and supply curves just reflect private costs and benefits. But in this question, since we are told it is a perfectly competitive market, we know that one of the assumptions is no externalities, so **that holds here and we can calculate it.**

Market quantity is 23, and total surplus is the area below the MWTP curve and above the MC curve up to the market quantity.

$$TS = 0.5 * 23 * (60 - 2.5) = 661.25$$

**Total surplus is \$661.25.**

3) A perfectly competitive market consists of 15 individual buyers and 10 individual sellers. Each buyer's demand curve is given by  $p = 30 - \frac{1}{4}q^D$ . Each supplier's supply curve is given by  $p = \frac{1}{4}q^S + 5$ . There are no externalities. (Notation: I'm using small letters for individuals and capital letters for markets to make things easier to keep track of visually. There is no deeper meaning to it.)

3a) Calculate market Consumer Surplus, market Producer Surplus and Total Surplus.

**Solution:** Step 1 is to calculate market supply and market demand. Recall that to do that we need to sum up individual *quantities* demanded and supplied at a given price. First, we need to rewrite each individual consumer's demand in terms of  $q^D(p)$  which gives us  $p = 30 - \frac{1}{4}q^D \Rightarrow q^D = 120 - 4p$ . Summing this up over 15 buyers gives us market demand

$$Q^D = 15 * (120 - 4p) = 1800 - 60p$$

Second, we need to rewrite each individual supplier's supply in terms of  $q^S(p)$  which gives us  $p = \frac{1}{4}q^S + 5 \Rightarrow q^S = 4p - 20$ . Summing this up over 10 sellers gives us market supply

$$Q^S = 10 * (4p - 20) = 40p - 200$$

To find market quantity and price we set

$$\begin{aligned} Q^D &= Q^S \\ 1800 - 60p &= 40p - 200 \\ P &= 20 \\ \Rightarrow Q &= 600 \end{aligned}$$

To calculate CS we need the intercept of the demand curve on the y-axis, which is 30. This makes CS the area of the triangle with base = 600 (the quantity), height = 30-20 (the difference between price and the intercept of the demand curve on the y-axis).

$$CS = 0.5 * (30 - 20) * 600 = 3000$$

Similarly, for PS we need the intercept of the supply curve on the y-axis, which is 5. This makes PS the area of the triangle with base = 600 (the quantity), height = 20-5 (the difference between price and the intercept of the supply curve on the y-axis).

$$PS = 0.5 * (20 - 5) * 600 = 4500$$

Since all benefits accrue only to the parties in the transaction and no one outside (i.e. no externalities), we can calculate total surplus using

$$TS = CS + PS = 7500$$

**Market CS is \$3,000, market PS is \$4,500, and total surplus is \$7,500.**

**3b)** Calculate each individual consumer and individual producer's surplus.

**Solution:** Use the individual buyer's (and seller's) demand (and supply) curve to calculate individual consumer (and producer) surplus.

To calculate individual CS we need the intercept of the demand curve on the y-axis, which is 30. This makes CS the area of the triangle with base = 40 (the individual quantity =  $600/15$ ), height =  $30-20$  (the difference between price and the intercept of the demand curve on the y-axis).

$$CS = 0.5 * (30 - 20) * 40 = 200$$

Similarly, for PS we need the intercept of the supply curve on the y-axis, which is 5. This makes PS the area of the triangle with base = 60 (the individual quantity =  $600/10$ ), height =  $20-5$  (the difference between price and the intercept of the supply curve on the y-axis).

$$PS = 0.5 * (20 - 5) * 60 = 450$$

Question: How is this related to the market numbers you calculated above? Why? **Each of the 15 consumers gets \$200 of surplus and each of the 10 producers gets \$450.**

**4)** Assume we have a perfectly competitive market with no externalities. The table below gives you total costs by quantity produced for two firms. The last column gives you the market willingness to pay (WTP) for each quantity. Assume that quantities are integers. *Note: As this table gives you only integer values, I should not have to say quantities must be only integers. It is the only possibility as you only have data on integers. With an equation, you have information for fractions of goods.*

| Q  | Firm 1<br>TC | Firm 2<br>TC | Market<br>WTP |
|----|--------------|--------------|---------------|
| 1  | 0.33         | 2            | 4.5           |
| 2  | 1            | 6            | 8.5           |
| 3  | 2            | 12           | 12            |
| 4  | 3.33         | 20           | 15            |
| 5  | 5            | 30           | 17.5          |
| 6  | 7            | 42           | 19.5          |
| 7  | 9.33         | 56           | 21            |
| 8  | 12           | 72           | 22            |
| 9  | 15           | 90           | 22.5          |
| 10 | 18.33        | 110          | 22.5          |
| 11 | 22           | 132          | 22            |
| 12 | 26           | 156          | 21            |

**4a)** What is the demand at a price of \$3?

**Solution:** The first step with any such problem is to calculate the Marginal WTPs, which are given in the table below. **At a price of \$3 the number of units that pass the  $MWTP \geq P$  threshold is 4.**

| Q  | MWTP |
|----|------|
| 1  | 4.5  |
| 2  | 4    |
| 3  | 3.5  |
| 4  | 3    |
| 5  | 2.5  |
| 6  | 2    |
| 7  | 1.5  |
| 8  | 1    |
| 9  | 0.5  |
| 10 | 0    |
| 11 | -0.5 |
| 12 | -1   |

4b) What is the supply at a price of \$3?

**Solution:** The marginal costs of the firms are given in the table below. **10 units pass the  $MC \leq P$  threshold: Firm 1 produces 9 and Firm 2 produces 1.**

| Q  | Firm 1<br>MC | Firm 2<br>MC |
|----|--------------|--------------|
| 1  | 0.33         | 2            |
| 2  | 0.67         | 4            |
| 3  | 1            | 6            |
| 4  | 1.33         | 8            |
| 5  | 1.67         | 10           |
| 6  | 2            | 12           |
| 7  | 2.33         | 14           |
| 8  | 2.67         | 16           |
| 9  | 3            | 18           |
| 10 | 3.33         | 20           |
| 11 | 3.67         | 22           |
| 12 | 4            | 24           |

4c) What quantity maximizes total surplus?

**Solution:** For this we need to find all the units for which the  $MWTP \geq MC$ . Lining up all the units in order of increasing MC and decreasing MWTP gives us the table below (incomplete, as not all the MCs have been included). **The quantity that maximizes total surplus is 6.** Notice that we do not need price to find total surplus or the allocation that maximizes it. That just depends on the difference between  $MWTP$  and  $MC$ . The price just determines how this surplus is split between consumers and producers, which for this question is irrelevant.

| Q  | MC   | MWTP |
|----|------|------|
| 1  | 0.33 | 4.5  |
| 2  | 0.67 | 4    |
| 3  | 1    | 3.5  |
| 4  | 1.33 | 3    |
| 5  | 1.67 | 2.5  |
| 6  | 2    | 2    |
| 7  | 2    | 1.5  |
| 8  | 2.33 | 1    |
| 9  | 2.67 | 0.5  |
| 10 | 3    | 0    |
| 11 | 3.33 | -0.5 |
| 12 | 3.67 | -1   |

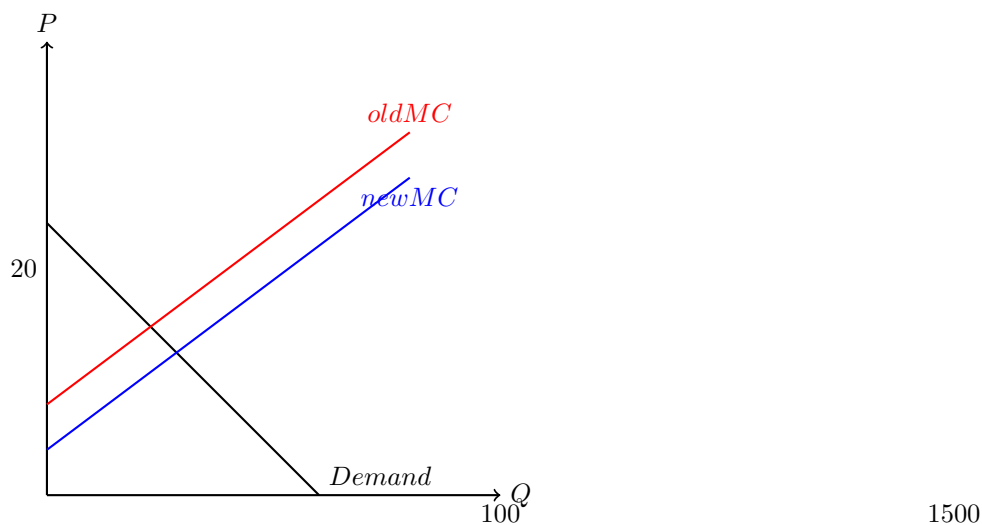
- 5) Assume we have a perfectly competitive market and no externalities. Remember we only have downward sloping demand and upward sloping supply in ECO101. Suppose that the marginal cost decreases by \$1 for each quantity produced. How does consumer surplus, producer surplus and total surplus change?

**Solution:** This is a parallel downward shift of the supply curve. As demand is downward sloping, we know that the change in market price is going to be less than \$1. See the diagram below to convince yourself. Market quantity bought and sold increases.

Consumer surplus increases by less than a dollar for every unit that was bought before (price is cheaper). In addition, there are new units bought and the additional CS from those units causes market CS to increase.

If you thought only about the old costs, you would say that producer surplus for previously sold units falls for the same reason that CS rises—producers are getting less money for those units than they did before. But remember that costs fall by \$1 and price falls by less than \$1. This means that PS increases for the previously sold units too. In addition, you have the increase in PS from the new units sold.

**An increase in both means that total surplus increases.**



- 6) The department of Economics introduces a new 4th-year course. More students want to take this course than there are spaces available. To deal with this they announce a lottery. Every student who wants to get in can sign up, and then they will draw names at random until all the spots are filled. Students can't exchange spots with each other. To help us think about the efficiency of this system let's use the example in this question.

We have 4 students whose willingness to pay for the course is in the table below. The lottery winners are Anisha and Nicolas.

|         | WTP    |
|---------|--------|
| Anisha  | \$3675 |
| Nicolas | \$6381 |
| Greta   | \$4523 |
| Oscar   | \$2391 |

6a) If Oscar had won a spot instead of Anisha what would have been the change in surplus?

**Solution:** If Anisha is made to give up her slot, surplus goes down by \$3675. When Oscar gets the spot, surplus goes up by \$2391. **The net change in surplus is -\$1284, i.e. surplus decreases by \$1284.**

6b) What would have been the change in surplus if the spaces had been allocated efficiently?

**Solution:** If the spaces had been allocated efficiently, they would have gone to the students with the highest WTP, which in this case are Nicolas and Greta. **The change in surplus from giving the spot to Greta and taking it away from Anisha is an increase of  $4523 - 3675 = \$848$ .**

6c) Suppose the department allows students to exchange spots with each other and doesn't ask too many questions. Is there a deal to be made here?

**Solution:** The problem is not the lottery but the fact that students can't exchange the spots and take advantage of gains from trade. In this case, Greta values it at \$4523 and Anisha values it at \$3675. There is space for a deal to be made. **For example, at any price  $\$3675 < p < \$4523$  both Anisha and Greta would be made better off.**

6d) Suppose instead that students on the list settle on a price of \$4000. If a student with a spot wants to sell a seat, they can sell it for \$4000 to a student who wants to buy one. Does this reach the efficient allocation? What is producer surplus from those who sell and what is the consumer surplus from those who buy?

**Solution:** At a price of \$4000, Nicolas' WTP is higher, so he keeps his spot. Anisha's is lower, so she sells hers. Greta's is higher, so she buys the spot, and Oscar's is lower, so he is not interested in purchasing one. **Consumer surplus is  $(4523 - 4000) = \$523$  from Greta, and producer surplus is  $(4000 - 3675) = \$325$  from Anisha.**

Notice the total change in surplus is \$848, the same as you found in part (b). This is another way you can see how prices work as a signal to move goods across buyers and sellers to reach the efficient allocation. People working in their own interest exhaust all the gains from trade by making mutually beneficial trades. The actual price doesn't matter for the allocation; it is just a way to determine how the surplus is shared between buyer and seller.

7) Consider a perfectly competitive market with our usual assumptions, including no externalities. Market Supply is given by  $Q^S = P$ . We have only two individuals in the market. Amanda's Marginal Willingness to Pay function is given by  $MWTP^A = 10 - Q$  and Bashir's is given by  $MWTP^B = 10 - 2Q$ .

7a) Calculate market price, Consumer Surplus (CS), Producer Surplus (PS) and Total Surplus (TS) in the market.

**Solution:** First calculate market demand by summing up individual demands. For Amanda we have  $MWTP = P = 10 - Q \Rightarrow Q = 10 - P$ . Similarly, for Bashir we get  $Q = 5 - \frac{P}{2}$ . Summing this up we get market demand as

$$Q^D = 15 - \frac{3P}{2}$$

Use Market supply to get

$$Q^D = 15 - \frac{3P}{2} = Q^S \Rightarrow Q = 6, P = 6$$

Consumer surplus is then

$$0.5 * (10 - 6) * 6 = \$12$$

Producer surplus is

$$0.5 * 6 * 6 = \$18$$

Total surplus is

$$TS = CS + PS = \$30$$

**Market price is \$6: CS is \$12, PS is \$18, and total surplus is \$30.**

- 7b) Suppose the market price is \$6. What is Amanda's CS and what is Bashir's CS? How is this related to the CS you calculated in part (a)?

**Solution:** Calculate individual CS from the individual demand curves.

**Amanda:**  $0.5 * (10 - 6) * 4 = \$8$

**Bashir:**  $0.5 * (10 - 6) * 2 = \$4$

**Market:** Amanda + Bashir = \$12, **the same market CS you calculated in part (a).** The lesson is that in a perfectly competitive market we can use market demand to work out market CS, and it gives the same answer as calculating each individual's CS and summing them. The same is true for producers.

- 8) Consider a perfectly competitive market with our usual assumptions, including no externalities. Market Supply is given by  $Q^S = 4P$ . Market demand is given by  $Q^D = 200 - P$ .

- 8a) Suppose that the quantity produced and exchanged in this market is 180. What is Total Surplus?

**Solution:** Don't worry too much about why the market quantity is what it is. You can assume for now that it is due to some government intervention. We will think about types of interventions in the Price Controls and Taxes topic.

I have purposely not given you a price, to drive home the insight that prices determine only the allocation of surplus. Revisit the lecture videos if this is still not clear or go to the discussion boards.

With the no externalities assumption, we have  $MSB = MWTP$  and  $MSC = MC$ . From the earlier modules we know that  $MWTP = P$  with the demand curve and  $MC = P$  with the supply curve.

To calculate the Total Benefit to society we find the area below the MSB (i.e. demand) curve for the first 180 units.

$$TSB = 0.5 * (200 - 20) * 180 + 20 * 180 = \$19800$$

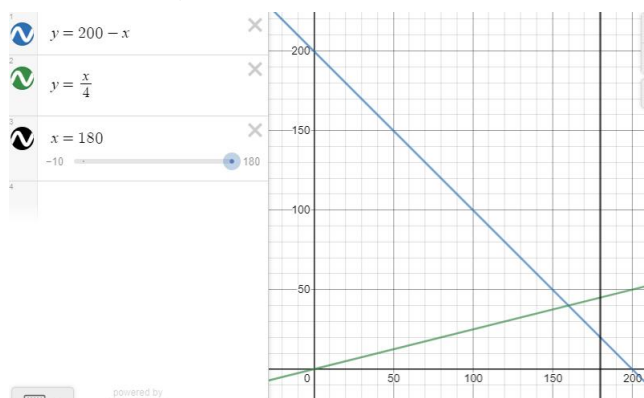
To calculate the Total Cost to society we find the area below the MSC (i.e. supply) curve for the first 180 units.

$$TSC = 0.5 * 180 * 45 = \$4050$$

Total Surplus is

$$TSB - TSC = \$15750$$

You can play around on Desmos to draw a diagram to see what is happening. **Total surplus at a quantity of 180 is \$15,750.**



Remember

1 We don't need prices to calculate total surplus. It is simply the area below the MSB and above the MSC for the units traded or, as we have done here, the total benefit to society from consuming the good minus the total cost to society from producing it.

2 CS, PS and TS are all denominated in dollars.

8b) What is the efficient quantity and what is Total Surplus at this quantity? Is there a loss in surplus (compared to the efficient) due to the restriction imposed above?

**Solution:** The efficient quantity is where  $MSB = MSC$ . With the no externalities assumption, we have  $MSB = MWTP$  and  $MSC = MC$ . From the earlier modules we know that  $MWTP = P$  with the demand curve and  $MC = P$  with the supply curve. To find the efficient quantity we find the quantity where the demand and supply curves intersect, giving

$$\begin{aligned}\frac{Q}{4} &= 200 - Q \\ Q &= 160 \\ TS &= 0.5 * (200 - 40) * 160 + 0.5 * 40 * 160 = \$16000\end{aligned}$$

Remember

1 We don't need prices to calculate total surplus. It is simply the area below the MWTP and above the MC for the units traded.

2 We don't have CS and PS here because there are no market prices. This is NOT the market allocation, the one in part (a) is the market allocation. This is the result of a thought exercise that says, IF we could choose, what quantity would we produce at?

3 IF we have a perfectly competitive market AND we did not have the intervention in part (a), THEN the market allocation would be this one and we would know the price (\$40 in this case).

**Yes**, there is a loss in efficiency due to the intervention in part (a), as the surplus is lower than the efficient one. **We can quantify this loss exactly as (16000-15750 =) a loss of \$250.**

**Why?** Because we are producing units (i.e. from 160 to 180) for which the cost to society is higher than the benefit to society. Producing them lowers the total surplus society enjoys.

9) Karen decides that as long as someone is willing to pay more than \$100, she'll sell her couch on Craigslist. She lists her couch for \$300, and the next day someone calls her up and offers her \$250. What is her Producer Surplus if she sells the couch?

**Solution: If Karen sells her couch, she will have a producer surplus of \$150.** Karen brings goods to the market, so she is a producer. PS is price minus willingness to sell =  $250 - 100 = \$150$ . It doesn't matter what she lists the couch for; the price she sells it for is the relevant price.

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Checked with Claude (Anthropic) for grammar, clarity and errors. Tell me if you find one I missed.